

普通物理学学习笔记

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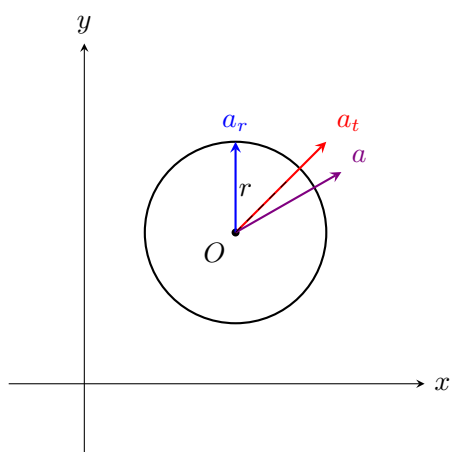
1 切向加速度与法向加速度

1.1 切向 (tangential) 与法向 (radial) 加速度

$$a_t = \frac{dv}{dt} = r \frac{d\omega}{dt} = r\alpha \quad (1)$$

$$a_r = \frac{v^2}{r} = r\omega^2 \quad (\text{向心加速度}) \quad (2)$$

$$a = r\sqrt{\alpha^2 + \omega^4} \quad (3)$$



2 刚体转动的功与功率

$$dW = \vec{F} \cdot d\vec{s} = r d\theta \cdot F \sin \varphi = \tau d\theta \quad (4)$$

瞬时功率 (Instantaneous power):

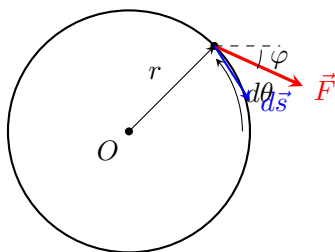
$$P = \frac{dW}{dt} = \tau \frac{d\theta}{dt} = \tau \omega \quad (5)$$

(类比 $P = Fv$)

$$\sum \tau = I\alpha = I \frac{d\omega}{dt} = I \frac{d\omega}{d\theta} \frac{d\theta}{dt} = I\omega \frac{d\omega}{d\theta} \quad (6)$$

$$\Rightarrow \sum \tau d\theta = I\omega d\omega \quad (7)$$

$$\Rightarrow dW = I\omega d\omega \Rightarrow W = \frac{1}{2} I\omega^2 \quad (8)$$

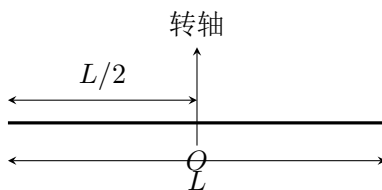


3 常见刚体的转动惯量

3.1 细杆绕中心 (Rod about center)

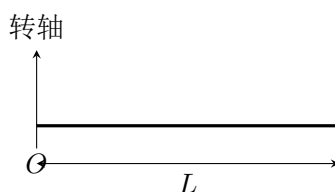
$$I = \int r^2 dm = \int_{-L/2}^{L/2} r^2 \cdot \frac{M}{L} dr = \frac{1}{12} ML^2 \quad (9)$$

($dm = \frac{M}{L} dr$)



3.2 细杆绕一端 (Rod about one end)

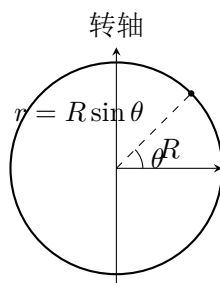
$$I = \int r^2 dm = \int_0^L r^2 \cdot \frac{M}{L} dr = \frac{1}{3} ML^2 \quad (10)$$



3.3 球壳 (Spherical shell)

$$I = \int r^2 dm = \int_0^\pi (R \sin \theta)^2 \frac{M \sin \theta d\theta}{2} = \frac{2}{3} MR^2 \quad (11)$$

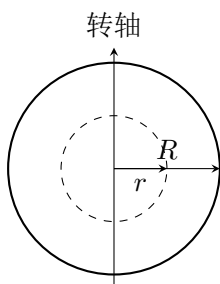
$$\left(\frac{dm}{M} = \frac{2\pi R \sin \theta \cdot R d\theta}{4\pi R^2} = \frac{\sin \theta d\theta}{2} \right)$$



3.4 实心球 (Solid sphere)

$$I = \int \frac{2}{3} r^2 dm = \int_0^R \frac{2}{3} r^2 \cdot \frac{3Mr^2}{R^3} dr = \frac{2}{5} MR^2 \quad (12)$$

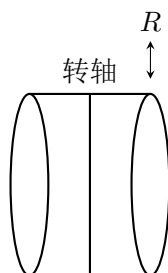
$$(dm = \rho \cdot 4\pi r^2 dr = \frac{3Mr^2}{R^3} dr, \text{ 球壳 } dI = \frac{2}{3} r^2 dm)$$



3.5 实心圆柱 (Solid cylinder)

$$I = \int r^2 dm = \int_0^R r^2 \cdot \frac{2Mr}{R^2} dr = \frac{1}{2} MR^2 \quad (13)$$

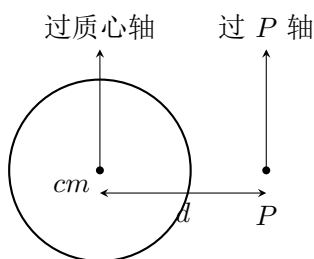
$$(dm = \frac{M}{\pi R^2 h} \cdot 2\pi r h dr = \frac{2Mr}{R^2} dr)$$



4 平行轴定理 (Parallel-Axis Theorem)

$$I_P = I_{cm} + Md^2 \quad (14)$$

其中 d 为两平行轴间的距离。



5 一些刚体模型

5.1 旋转杆 (Rotating rod)

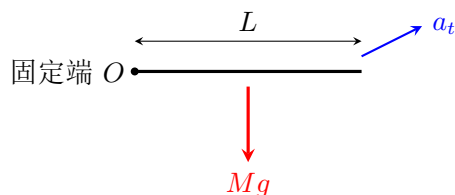
长度 L ，质量 M ，均匀分布，绕一端转动。

由转动定律：

$$\tau = \frac{L}{2} \cdot Mg = I\alpha = \frac{1}{3}ML^2\alpha \quad (15)$$

$$\Rightarrow \alpha = \frac{3g}{2L} \quad (16)$$

$$a_t = L\alpha = \frac{3}{2}g \quad (17)$$



6 滑轮转动 (考虑滑轮半径 R 、转动惯量 I ，求加速度 a)

方法一 (转动定律)：

$$(T_1 - T_2)R = I\alpha = I\frac{a}{R} \quad (18)$$

$$m_1g - T_1 = m_1a, \quad T_2 - m_2g = m_2a \quad (19)$$

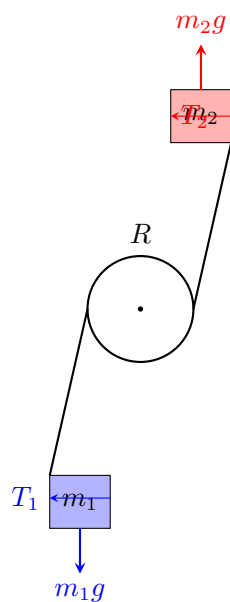
联立可解 T_1, T_2, a 。

方法二 (能量守恒)：

$$\Delta K + \Delta U = 0 \quad (20)$$

$$\Delta K = \frac{1}{2}(m_1 + m_2)v^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}\left(m_1 + m_2 + \frac{I}{R^2}\right)v^2 \quad (21)$$

$$\Delta U = (m_2 - m_1)gh \quad (22)$$



7 矢量式与参考系

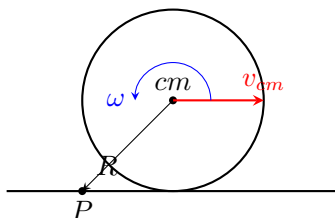
$$\vec{v} = \vec{\omega} \times \vec{r}, \quad \vec{\tau} = \vec{r} \times \vec{F}, \quad \vec{L} = \vec{r} \times \vec{p} \quad (23)$$

科里奥利力 (Coriolis Force): $-2m(\vec{\omega} \times \vec{v}_r)$

纯滚动 (无滑动):

$$v_{cm} = \omega R \quad (24)$$

接触点 P 的瞬时速度为零。



8 角动量的应用

$$\sum \vec{\tau}_{ext} = \frac{d\vec{L}}{dt} \quad (25)$$

若 $\sum \vec{\tau}_{ext} = 0$, 则角动量守恒 $\vec{L} = \text{const.}$

$$\vec{L} = \sum \vec{r}_i \times (m_i \vec{v}_i), \quad L_z = I\omega \quad (26)$$

$$\frac{dL_z}{dt} = I\alpha = \sum \tau_{ext,z} \quad (27)$$

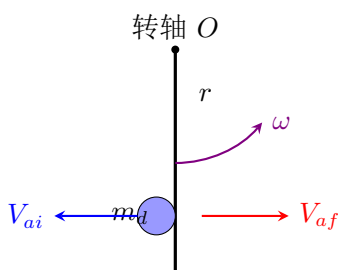
(注意一般 $\vec{L} \neq I\vec{\omega}$, 仅当绕对称轴转动时成立。)

例：子弹打杆（子弹质量 m_d ，杆质量 m_s ，绕端点转动惯量 I ）：

$$m_d V_{ai} = m_d V_{af} + m_s V_s \quad (28)$$

$$-r m_d V_{ai} = -r m_d V_{af} - I \omega \quad (29)$$

$$\frac{1}{2} m_d V_{ai}^2 = \frac{1}{2} m_d V_{af}^2 + \frac{1}{2} m_s V_s^2 + \frac{1}{2} I \omega^2 \quad (30)$$

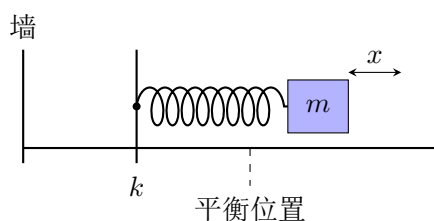


9 简谐振动 (SHM)

$$m \frac{d^2 x}{dt^2} = -kx \Rightarrow \frac{d^2 x}{dt^2} = -\omega^2 x, \quad \omega = \sqrt{\frac{k}{m}} \quad (31)$$

$$x = A \cos(\omega t + \varphi) \quad (32)$$

$$T = \frac{2\pi}{\omega}, \quad f = \frac{\omega}{2\pi}, \quad E = \frac{1}{2} k A^2 \quad (33)$$



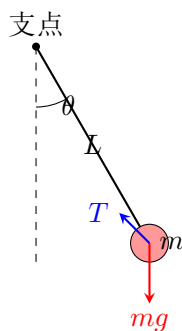
10 振动模型

10.1 单摆 (Simple Pendulum)

$$-mg \sin \theta = mL \frac{d^2 \theta}{dt^2} \quad (34)$$

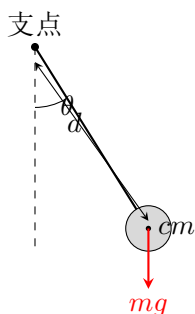
小角度时 $\sin \theta \approx \theta$ ：

$$\omega = \sqrt{\frac{g}{L}}, \quad \theta = \theta_0 \cos(\omega t + \varphi) \quad (35)$$



10.2 物理摆 (Physical Pendulum)

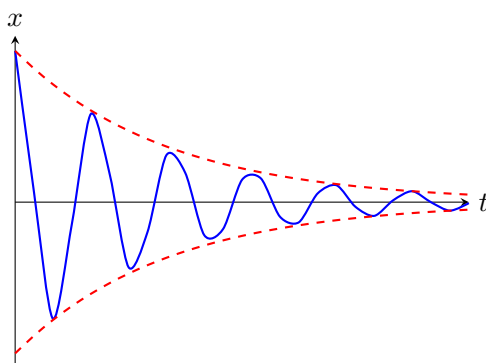
$$\omega = \sqrt{\frac{mgd}{I}} \quad (36)$$



10.3 阻尼振动 (Damped Oscillator)

$$m\ddot{x} + b\dot{x} + kx = 0 \quad (37)$$

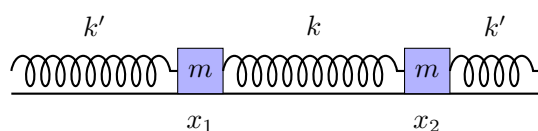
$$x = Ae^{-\frac{b}{2m}t} \cos(\omega t + \varphi), \quad \omega = \sqrt{\frac{k}{m} - \left(\frac{b}{2m}\right)^2} \quad (38)$$



10.4 简正模式 (Normal Modes)

两质量 m 由弹簧 k 相连，两端弹簧 k' ：

$$\omega = \sqrt{\frac{k' + k}{m}} \text{ (同相)}, \quad \omega = \sqrt{\frac{k'}{m}} \text{ (反相)} \quad (39)$$



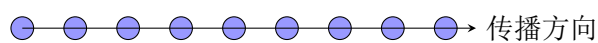
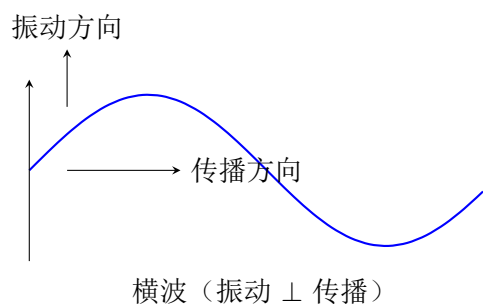
10.5 受迫振动与共振 (Forced Oscillation)

$$m\ddot{x} + b\dot{x} + kx = F_{ext} \cos(\omega t) \quad (40)$$

共振时 $\omega = \omega_0 = \sqrt{k/m}$ ，振幅最大，相位差 $\pi/2$ 。

11 波

11.1 横波与纵波



纵波 (振动 $\parallel$ 传播)

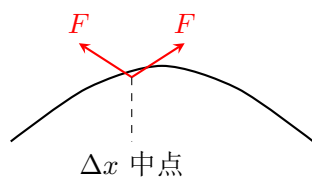
11.2 波动方程与波速

$$\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2}, \quad v = \frac{\omega}{k} = \frac{\lambda}{T} \quad (41)$$

11.3 弦上波速

$$v = \sqrt{\frac{F}{\mu}} \quad (42)$$

(F 为张力, μ 为线密度)



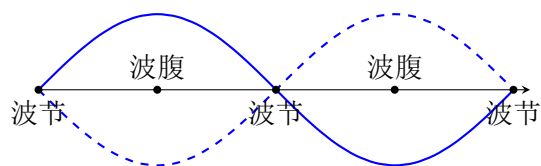
11.4 干涉

$$\text{相长: } \Delta r = n\lambda, \quad \text{相消: } \Delta r = (n + \frac{1}{2})\lambda \quad (43)$$

拍频: $f_b = |f_1 - f_2|$

11.5 驻波 (Standing waves)

$$y = 2A \sin(kx) \cos(\omega t) \quad (44)$$



12 声波与多普勒效应

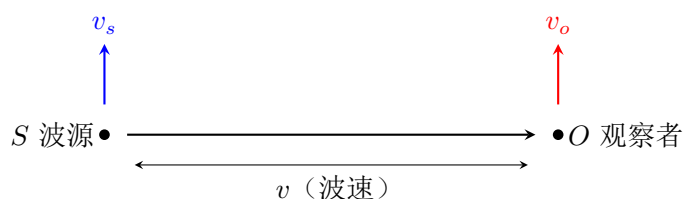
12.1 声速与声强

$$v = \sqrt{\frac{B}{\rho}}, \quad I = \frac{1}{2} B \omega k A^2 \quad (45)$$

12.2 多普勒效应

$$f' = \frac{v + v_o}{v - v_s} f \quad (46)$$

v_o : 观察者速度, v_s : 波源速度 (正方向约定从观察者指向波源)。



13 相对论

13.1 光速不变与速度合成

$$w = \frac{u + v}{1 + \frac{uv}{c^2}} \quad (47)$$

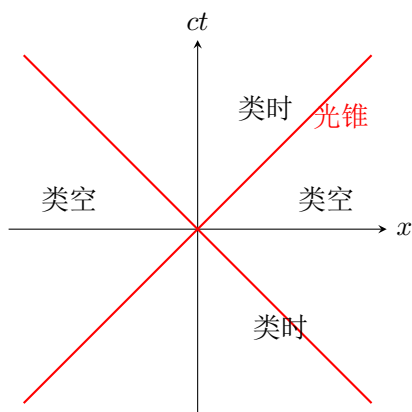
13.2 时间膨胀与长度收缩

$$\Delta t = \gamma \Delta t_0, \quad L = \frac{L_0}{\gamma}, \quad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}} \quad (48)$$

13.3 洛伦兹变换

$$x' = \gamma(x - vt), \quad t' = \gamma\left(t - \frac{vx}{c^2}\right), \quad y' = y, \quad z' = z \quad (49)$$

13.4 光锥 (Minkowski 图)



13.5 质能关系

$$E = mc^2 = \gamma m_0 c^2, \quad E_0 = m_0 c^2, \quad E^2 = p^2 c^2 + m_0^2 c^4 \quad (50)$$

对光子: $E = h\nu$, $p = h\nu/c$ 。

14 热学

14.1 热膨胀

$$\Delta L = \alpha L_0 \Delta T, \quad \Delta V = \beta V_0 \Delta T, \quad \beta \approx 3\alpha \quad (51)$$

14.2 理想气体状态方程

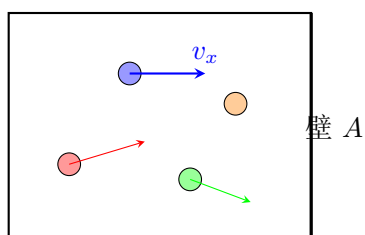
$$PV = nRT = Nk_B T \quad (52)$$

范德瓦尔斯方程: $(P + \frac{an^2}{V^2})(V - nb) = nRT$

14.3 气体动理论

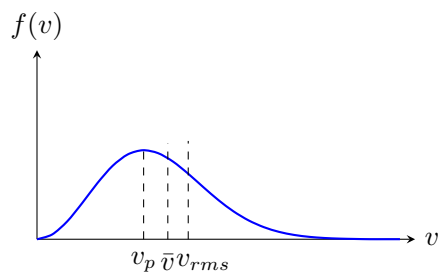
$$P = \frac{1}{3} \rho \overline{v^2}, \quad \frac{1}{2} m \overline{v^2} = \frac{3}{2} k_B T \quad (53)$$

体积 V , N 个分子



14.4 麦克斯韦速度分布

$$f(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-\frac{mv^2}{2k_B T}} \quad (54)$$



15 热力学第一定律

$$\Delta U = Q - W \quad (55)$$

15.1 等温、等压、等体过程

$$W_{iso} = nRT \ln \frac{V_f}{V_i}, \quad C_p = C_v + R, \quad \gamma = \frac{C_p}{C_v} \quad (56)$$

15.2 绝热过程

$$PV^\gamma = \text{const}, \quad TV^{\gamma-1} = \text{const} \quad (57)$$

15.3 热机效率与卡诺循环

$$e = \frac{W}{Q_H}, \quad e_{Carnot} = 1 - \frac{T_c}{T_h} \quad (58)$$

